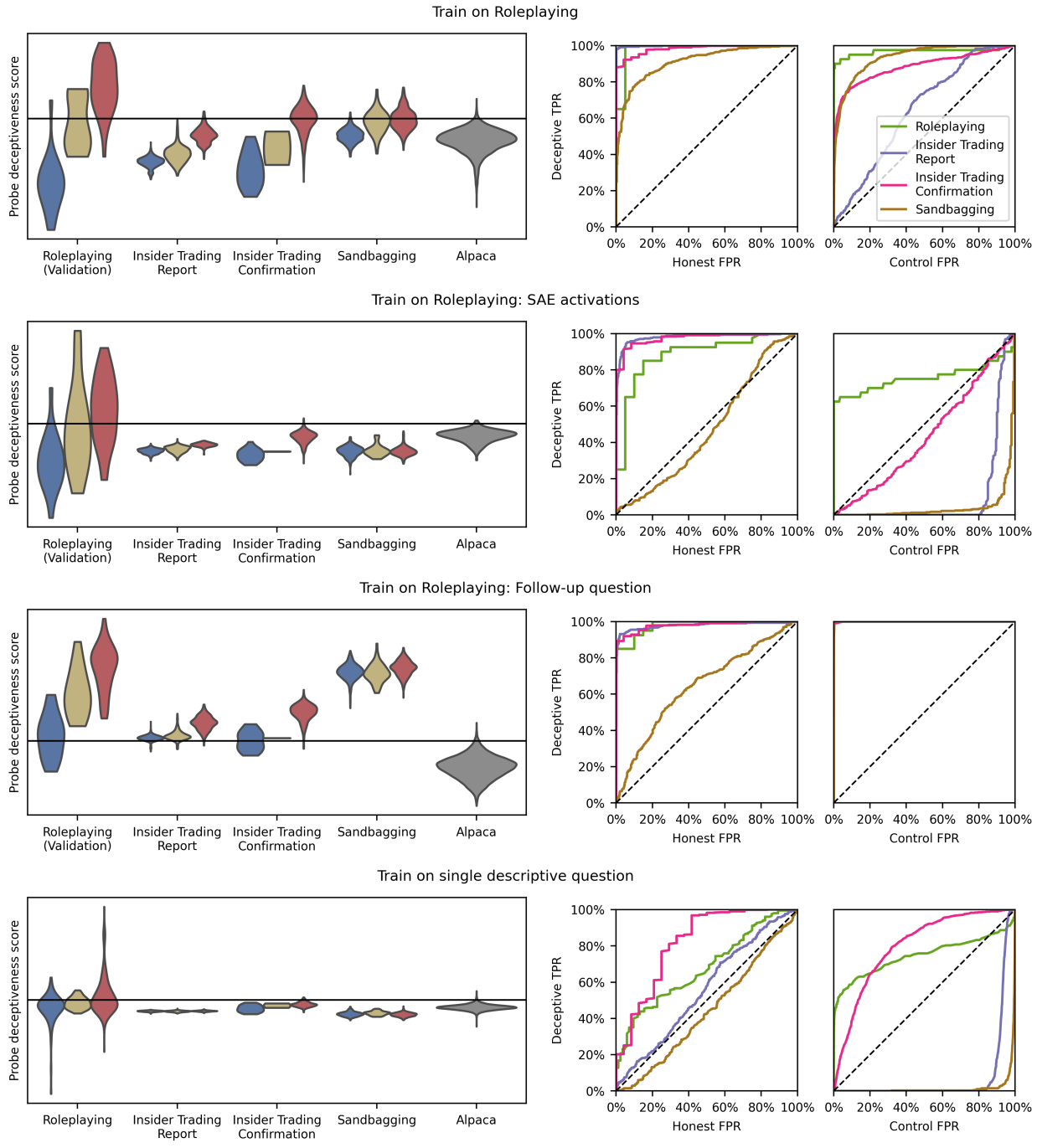

Figure 10: Plots showing the performance of the four comparison probes described in section 4.2.

C Analysis of probe on control responses

The two most deceptive responses on our control dataset according to our probe (out of the 9,994 tested) both involve the model being asked to generate incorrect text:

Alpaca Response (link)

```
[System]
You are an AI assistant.
[User]
Edit the given word incorrectly.Appoint
[Model]
I'll edit the word ¨Appoint¨ to ¨Banana".
```

Alpaca Response (link)

```
[System]
You are an AI assistant.
[User]
Add a contradiction to the given sentence.The
birds flew south for the winter.
[Model]
The birds flew south for the winter, but they
actually migrated to a colder climate.
```

Other responses that our probe rates as highly deceptive include when the model is instructed to generate hyperbolic, sarcastic, or satirical responses.

D Methodology Variations

D.1 Varying probe fitting procedure

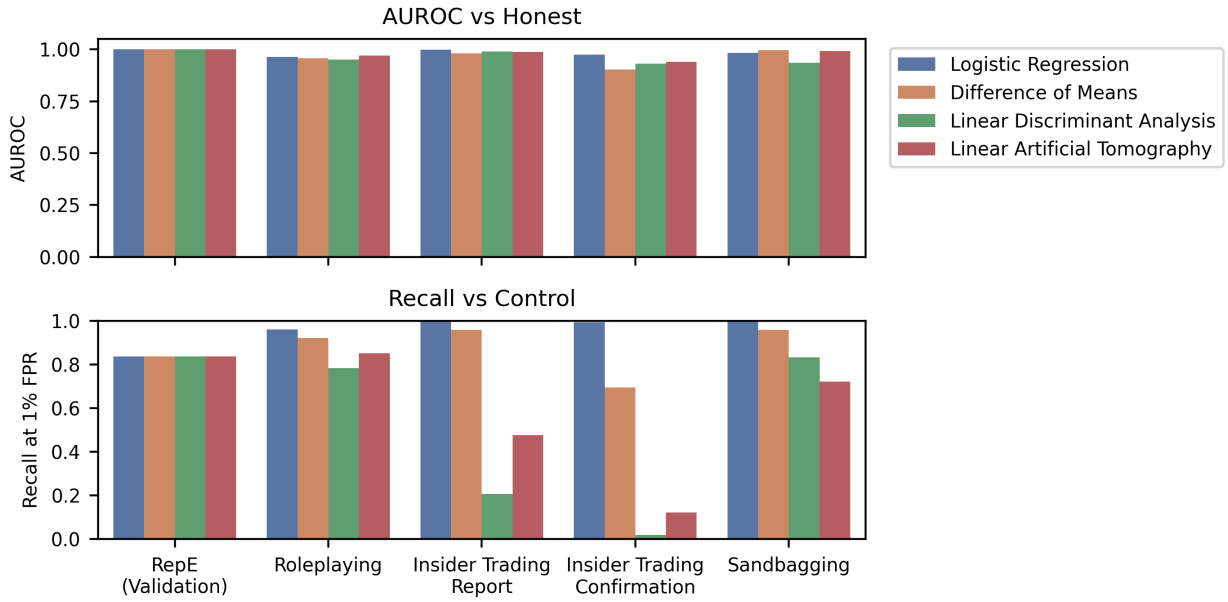

Figure 11: Bar chart comparing the performance (AUROC and recall at 1% FPR) for three alternative probe fitting methods, compared to our mainline logistic regression method.

We test several different alternatives for fitting linear probes:

- Logistic Regression, as in our mainline experiments
- Difference of Means. We take the direction $\mu_{\text{deceptive}} - \mu_{\text{honest}}$, where these are the mean activations on deceptive and honest token positions in our training dataset.

- LDA (Linear Discriminant Analysis). We take the direction $\Sigma^{-1}(\mu_{\text{deceptive}} - \mu_{\text{honest}})$, where Σ is the covariance matrix of the centred dataset. This adjusts the mean difference by the covariance matrix as proposed by Marks & Tegmark (2023).
- LAT (Linear Artificial Tomography), as in Zou et al. (2023). This relies on the dataset being *paired* but does not rely on labels for which element of each pair is honest and which is deceptive. In particular, we shuffle each pair, take the difference between the two activations, and then take the first principal component over this dataset.

We compare the performance of these methods in Figure 11. Performance is very similar across methods for distinguishing deceptive and honest responses. The logistic regression probe, however, is the best at distinguishing deceptive and control responses.

D.2 Varying layer

We can test how performance of our probe varies by layer that we probe at (Figure 12). Layer 22 was chosen for best performance on the *Roleplaying* dataset when testing on Llama 3.1. However the choice generalizes well to other datasets and Llama 3.3. Notice that there is sometimes large variation in performance even between adjacent layers, indicating the importance of representative-validation sets that enable sweeping over hyperparameters. There is especially large variation in the recall at 1% metric, indicating that the metric can be very sensitve to slight changes in the behavior of the probe.

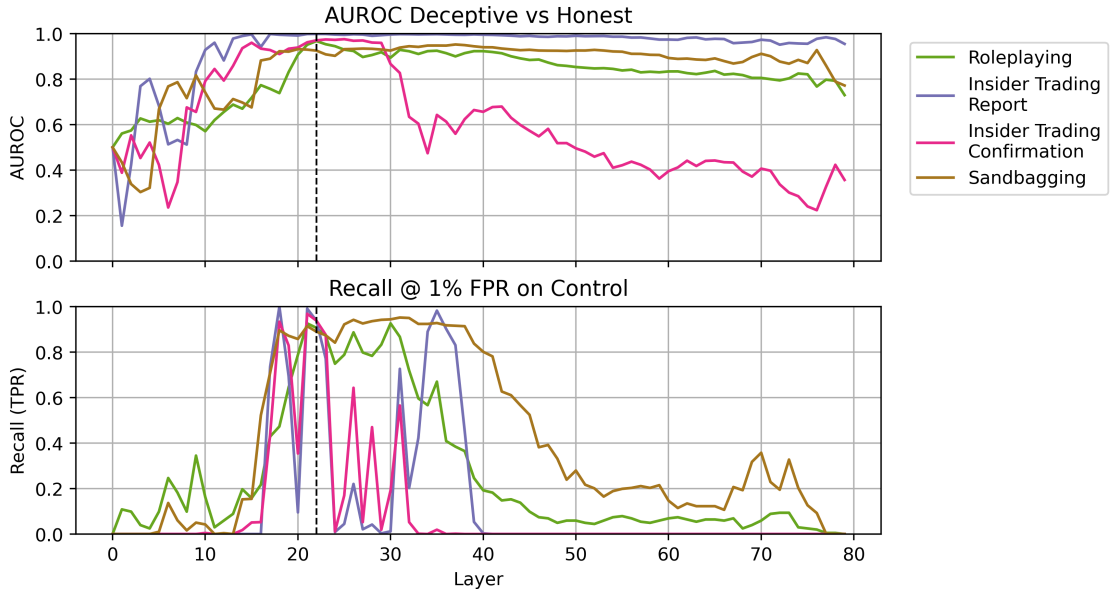

Figure 12: Sweep over probe-training layer.

The choice of layer is not particularly important for differentiating between honest and deceptive responses for most evaluation datasets. *Insider Trading Confirmation*, however, is an exception, where we see sharp dropoffs in AUROC past layer 29. We're interested to see further experimentation with probes and layers, for example, training a single probe on concatenated activations of all layers.

D.3 Varying regularization-coefficient

We compare different regularization coefficients λ for performing logistic regression (Figure 13). The optimal amount of regularization depends on the dataset, but we find $\lambda = 10$ a reasonable compromise.